

SIQI LI

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Ph.D. student building vision-language models, real-time VR perception, and embodied AI systems.

EDUCATION

University of California, Irvine

Ph.D. in Computer Engineering GPA: 4.0/4.0

Advisor: Prof. Yasser Shoukry | Co-advisor: Prof. Unnat Jain

Irvine, CA, USA

Expected Aug 2028

Karlsruhe Institute of Technology

M.Sc. in Mechatronics and Information Technology (Robotics)

Karlsruhe, Germany

Tongji University

B.Eng. in Vehicle Engineering

Shanghai, China

TECHNICAL SKILLS

Multimodal ML	VLM fine-tuning, vision-language-action models, diffusion transformers, CLIP, PyTorch, Hugging Face
Vision and VR	Real-time object detection, YOLO, in-headset privacy masking, VR perception, ONNX, ONNX Runtime, TensorRT
Simulation and Control	Simulation-driven evaluation, embodied agents, dynamic manipulation, model predictive control, trajectory optimization, ROS
LLMs and Evaluation	Preference alignment, personalization, retrieval-augmented generation, adversarial detection, evaluation pipelines
Programming and Tools	Python, C/C++, MATLAB, Julia, W&B, AWS

RESEARCH EXPERIENCE

Vision-Language-Action Models for Dynamic Manipulation

University of California, Irvine | Ph.D. Student

Sep 2024–Present

- Fine-tuned a pretrained **vision-language model (VLM)** backbone and **diffusion transformer (DiT)** action expert for a velocity-aware VLA model that jointly predicts robot joint positions and velocities.
- Designed a controller combining position and velocity predictions; built tossing, sliding, and flipping evaluation tasks to assess dynamic manipulation.

Real-Time VR Perception and Robust Computer Vision

University of California, Irvine | Ph.D. Student

Sep 2024–Present

- Deployed **YOLO via ONNX** for **real-time, in-headset object detection and privacy masking** on a VR rehabilitation platform with UC Berkeley and UCSF.
- Developed KoALA, an adversarial detector combining KL divergence and L_0 prediction agreement with a formal detection guarantee; evaluated ResNet-18 and the **CLIP ViT-B/32** vision-language model.

Language-Guided Robot Planning with VLM Feedback

California Institute of Technology | Visiting Student

Dec 2022–May 2024

- Integrated an LLM with model predictive control for language-guided robot planning and motion control; incorporated **VLM feedback** into the control framework.

LLM Preference Alignment and Personalization

University of California, Irvine | Ph.D. Student

Sep 2024–Present

- Built a persona-conditioned framework and **1,200-example dataset**, increasing preference-following accuracy from **29.25% to 76%** (+46.75 percentage points); developed a RAG-based evaluation pipeline.

Simulation-Based Planning and Retrieval-Augmented Speech Translation

Karlsruhe Institute of Technology | Research Assistant / Master's Thesis

Jul 2021–Nov 2022

- Developed nonlinear optimization and rule-based lane-change planners with **vehicle simulations**; built retrieval-augmented speech translation targeting rare-word accuracy (EMNLP 2024).

Energy-Efficient Vehicle Control and Privacy-Preserving Computation

Tsinghua University | Research Assistant

Jul 2020–Nov 2020

- Developed energy-saving vehicle control using dynamic programming and MPC; contributed to road testing and privacy-preserving cloud computation.

INDUSTRY EXPERIENCE

Brose-China

Software Engineer Intern

Apr 2019–Aug 2019

- [1] Siqi Li*, Zhenyang Feng, Unnat Jain, Yasser Shoukry.
Velocity Matters: Coupled Position–Velocity Prediction in Vision-Language-Action Models for Dynamic Robot Manipulation.
Manuscript in preparation for ICLR 2027.
- [2] Gina Gwiazda, Alison Scheid, Edward Kim, Yuri Cho, Jeremy Fischer, Siqi Li, Mahmoud Elfar, Erik Nelson, Seonghyun Park, Rachel Zeng, Kamesh Krishnamurthy, Jessica Mok, Yasser Shoukry, Bjoern Hartmann, Sanjit A. Seshia, Cathra Halabi.
Human-centered Design of a Privacy-preserving Mixed-reality Exercise Program for Stroke (P5-14.006).
Neurology, 106(11_Supplement_1):3130, 2026.
- [3] Siqi Li*, Vy Ngoc Huynh, Mahmoud Elfar, Edward Kim, Bjoern Hartmann, Sanjit A. Seshia, Yasser Shoukry.
Look Away: Steering User Attention to Protect Privacy in Extended Reality Environments.
SOUPS 2026 — under review.
- [4] Tianyu Zhao*, Siqi Li*, Yasser Shoukry, Salma Elmalaki.
Can LLMs Discern the Traits Influencing Your Preferences? Evaluating Personality-Driven Preference Alignment in LLMs.
arXiv:2602.07181 (2026). Lifelong Agents Workshop @ COLM 2026 — accepted.
- [5] Siqi Li*, Yasser Shoukry.
KoALA: KL-L0 Adversarial Detector via Label Agreement.
arXiv:2510.12752 (2025). Submitted to AAAI 2027.
- [6] Siqi Li*, Danni Liu*, Jan Niehues.
Optimizing Rare Word Accuracy in Direct Speech Translation with a Retrieval-and-Demonstration Approach.
EMNLP 2024. DOI: 10.18653/v1/2024.emnlp-main.708.
- [7] Yan Cui*, Siqi Li*, Yue Wang, Bolin Gao.
The Data Protection of Intelligent Connected Vehicles Cloud Control Framework Using Fully Homomorphic Encryption.
2020 4th CAA International Conference on Vehicular Control and Intelligence (CVCI), pp. 19–24. DOI: 10.1109/CVCI51460.2020.9338620.

PATENT

Keqiang Li*, Siqi Li*, Bolin Gao, Yan Cui, Yue Wang.
Cloud Control Vehicle Energy-Saving System with Data Encryption Method Applied in Intelligent Connected Vehicle (ICV).
CN202011500343.6[P].

* Equal contribution.